

Augmented Sample Size Calculation With Truncated Data

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Incomplete data, including truncated and censored observations, can lead to information loss and affect the accuracy of statistical inference. This thesis develops an effective sample size framework (ESS) to quantify information loss in incomplete data. An Expectation–Maximization (EM) algorithm is used to estimate the ESS, which is then incorporated into inference procedures by replacing the nominal sample size.

To formalize this idea, let O denote the set of fully observed observations and C denote the set of censored observations. The effective sample size is defined as Combining fully observed, truncated, and censored observations, the effective sample size can be written as

$$m_{\text{eff}} = \sum_{i \in O} 1 + \sum_{j \in T} w_j^{(T)} + \sum_{k \in C} w_k^{(C)}, \quad (1)$$

where O , T , and C denote the sets of fully observed, truncated, and censored observations, respectively. For interval-truncated observations,

$$w_j^{(T)} = 1 - \left[\Phi \left(\frac{U_j - \hat{\mu}}{\hat{\sigma}} \right) - \Phi \left(\frac{L_j - \hat{\mu}}{\hat{\sigma}} \right) \right], \quad (2)$$

while for right-censored observations,

$$w_k^{(C)} = 1 - P(X_k > C_k) = \Phi \left(\frac{C_k - \hat{\mu}}{\hat{\sigma}} \right). \quad (3)$$

Simulation studies compare the proposed method with conventional approaches using confidence interval coverage, Type I error, power, bias, and mean squared error. survival times from the Heart Failure Clinical Records dataset in UC Irvine as real-data applications are also considered. The results show that the proposed ESS effectively reflects information loss and generally provides more reliable inference than methods based only on the observed sample size. In particular, censored data exhibit greater information loss than truncated data, leading to a substantially smaller ESS.

References

- [1] M.R. Munasinghe, *Robust Regression Models with Missing and Censored Data via the EM Algorithm*, Ph.D. thesis, Thompson Rivers University (2023).
- [2] T. Liu, J. Fan, Y. Luo, N. Tang, G. Li, and X. Du, *Adaptive Data Augmentation for Supervised Learning over Missing Data*, in *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data*, ACM, pp. 1757–1770 (2020). doi: <https://doi.org/10.1145/3318464.3389736>.
- [3] Y. Bian, G.Y. Yi, and W. He, *Boosting Methods for Interval-Censored Data with Regression and Classification*, (2026).
- [4] X. Chen, Y. Liu, and T. Sun, *Constrained Optimization Methods in Modern Statistical Learning*, *Journal of Machine Learning Research* **22**, pp. 1–45 (2021).
- [5] S. Barman, R. Krishnaswamy, and X. Wu, *Optimization with Constraints: Theory and Applications*, *Foundations and Trends in Theoretical Computer Science* **13**, 4, pp. 247–376 (2018). doi: <https://doi.org/10.1561/04000000078>.